Understanding ACT: The CVAE Encoder Phase

Explaining the Style Variable Generation

1 Introduction

During the training phase of Action Chunking with Transformers (ACT), the model behaves as a Conditional Variational Autoencoder (CVAE). This means before the main Transformer (the Decoder) can process the vision and proprioception context, we must compute a "style variable," denoted as z. This variable is designed to capture the stochasticity, multi-modality, and specific human nuances of a given demonstration.

This document details the pipeline of the CVAE Encoder up to the point where all arguments are prepared for the main loss function.

2 The CVAE Encoder Inputs

To calculate z during training, the CVAE Encoder is fed a combination of the current state and the actual future actions from the dataset. Interestingly, to save computation, the heavy image features are **not** fed to the encoder.

- **Proprioception (o_t):** The normalized joint positions of the single-arm state.
- **Future Actions ($a_{t:t+k}$):** The sequence of normalized k future target joint positions that the human expert actually executed.
- **[CLS] Token:** A learnable class token (similar to the one used in BERT) which is prepended to the input sequence.

3 The Encoder Forward Pass

The inputs are mapped to the hidden dimension size (e.g., 512) and passed through a BERT-like Transformer Encoder.

3.1 Aggregating Information

Inside this Encoder, self-attention layers allow the `[CLS]` token to interact with the entire future action sequence and the proprioception vector. By the time it passes through all the layers, the final output corresponding to the `[CLS]` position acts as a summarized mathematical representation of the entire human action sequence's "style" in that specific state.

3.2 Predicting the Distribution

Instead of generating a static tensor, the summary `[CLS]` feature is passed through linear layers to predict a statistical distribution, defined by two vectors:

1. **Mean (μ):** The center of the "style" distribution.
2. **Standard Deviation (σ or $\log(\sigma^2)$):** The spread or uncertainty of the distribution.

4 Sampling and the Reparameterization Trick

Instead of directly passing the mean as z, we sample from the predicted Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$.

To maintain differentiability for backpropagation during training, a concept called the *reparameterization trick* is applied:

$$z = \mu + \sigma \cdot \epsilon$$

Where ϵ is random noise sampled from a standard normal distribution $\mathcal{N}(0, 1)$. This z vector is finally projected to the standard hidden dimension (e.g., 512).

5 Preparing for the Loss Function

At this point, we have successfully generated everything needed to fulfill the two components of the CVAE loss function:

- **KL Divergence Arguments:** We now have the predicted parameters (μ, σ) from the CVAE Encoder. These are the exact arguments needed to compute the Kullback-Leibler (KL) Divergence loss. This loss component will mathematically penalize the model if the predicted distribution (μ, σ) strays too far from a standard normal prior distribution $\mathcal{N}(0, I)$.
- **Reconstruction Arguments:** We have the sampled style vector z. This z is combined with the *live* inputs (the 600-900 visual tokens and the single state token derived in the previous phase) to act as the complete context for the main Transformer Decoder. The Decoder will use this context to generate a predicted chunk of k actions. These predictions, alongside the ground truth actions $a_{t:t+k}$, represent the final arguments needed to compute the standard L1 Reconstruction Loss.

By finalizing the generation of μ, σ, and z, the encoder's job is done. The inputs are staged, the style is defined, and both terms of the CVAE loss function are fully prepped for computation.